

Projects: EUGLOH workshop on the Physics for Biology in Paris, June 2026

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Brownian motion in the body

NOTE: * denotes extra tasks to be considered if time permits!

Part A: Simulating Brownian motion

As seen in the lecture notes, diffusion is an essential transport mechanism over cellular length scales. Often-times, the Brownian motion of proteins is studied under a camera-microscopy setup. The goal is then to estimate the proteins' diffusion constants from the experimental movies. How can one estimate the diffusion constant from such measured particle trajectories?

Here, you will simulate Brownian motion trajectories and use statistical methods to estimate the diffusion constant. Consider the one-dimensional stochastic process

$$x_{i+1} = x_i + \sqrt{2D\Delta t} \eta_i$$

where D is the diffusion constant, Δt is the time step, and η_i are independent Gaussian random variables with mean zero and variance one.

Tasks

1. Simulate trajectories for different diffusion constants.
 - Estimate D from the size of typical proteins, which you can look up. Hint: use the Stokes-Einstein relation from the lecture notes.
 - The time step is set by the imaging camera's exposure time. Look up a reasonable value of Δt for a camera designed for fluorescence microscopy.
 - Draw Gaussian random variables using built-in libraries. (If there was a closed formula for its CDF one could have implemented the transformation method in section 2.4.2 of the lecture notes).
2. Plot a few sample trajectories as function of time.
 - For convenience, use units of microns and seconds.
 - Suppose we could track the molecules at a much higher time resolution. What happens qualitatively if Δt is made much smaller while holding the total simulation time fixed?

3. Compute the mean-squared displacement (MSD)

$$\langle (x(t) - x(0))^2 \rangle$$

and verify the diffusion law

$$\langle (x(t) - x(0))^2 \rangle = 2Dt.$$

See section 3.3 in the lecture notes.

4. Check how good an estimate of D you get for trajectories of a certain length or number.
5. * When localising the particles in fluorescence microscopy, small errors occur, i.e., the measured positions of the particle are not exact.
 - Read up on localization noise in the following article: Vestergaard, C. L., Blainey, P. C., & Flyvbjerg, H. (2014). Optimal estimation of diffusion coefficients from single-particle trajectories. *Physical Review E*, 89, 022726.
 - Trajectories that have localization errors are easy to simulate: to each particle position add a Gaussian random number with variance σ^2 . Assume that $\sigma = 20$ nm.
 - In the article, zoom in to Eqs. 14 and 15, which give explicit expressions for estimator for D and the localization error. Use these and compare to the ground truth values.

Part B: Estimating the diffusion constant

In part A, we used a simple estimator for the diffusion constant. In any type of estimation however, an accompanying confidence interval will be asked for. The most complete such is a probability distribution over the most likely value of D from the analysed trajectories. Here you will use Bayes theorem to find the posterior distribution. Note that, whereas in part A you found the estimation error by comparing to the simulated value of D , here you will not need to.

The probability density for a Brownian displacement during a time interval Δt is

$$p(\Delta x|D) = \frac{1}{\sqrt{4\pi D\Delta t}} \exp\left(-\frac{(\Delta x)^2}{4D\Delta t}\right).$$

For an entire trajectory we get the product

$$\mathcal{L}(D) = \prod_i p(\Delta x_i|D)$$

where

$$\Delta x_i = x_{i+1} - x_i.$$

Tasks

1. Suppose you knew an upper and a lower limit for the true value of D , before analysing any data. What is the corresponding uniform distribution? How could this encode your prior

knowledge of D ?

2. From a set of simulated trajectories, compute

$$P(D|\text{data}) \propto P(\text{data}|D)P(D)$$

and plot it as a function of D .

- How can we turn this into a probability distribution? Do this.
 - What does the distribution look like and what does it mean? Do you expect the distribution to become wider or thinner as you analyse longer/more trajectories?
 - How would you go about to compute confidence intervals from this distribution? (You don't need to do it).
3. Simulate lots of data sets and for each one, compute point estimates of D via the estimator in part A. Where do they fall on the posterior distribution for D ? (You may either plot the estimates directly on top of the curve, or make a histogram of the point estimates). Try to explain your findings in words.
 4. * What other prior distributions could be used here? Do they contain more or less information than the uniform prior? Do a quick literature study and in the end double check that the proposed distributions make sense for $D \in (0, \infty)$.

Part C: First-passage times

Suppose a ligand-receptor reaction needs to take place in a cell. Within such small distances, a valid strategy of the receptor is simply to wait for the ligand to show up. Let diffusion do the work! The only caveat is that the transport is ruled by probabilities. So how long does the receptor need to wait for the ligand?

Here you will study how long it takes for a diffusing particle to find a target. Consider a random walker in one dimension and suppose there is an absorbing target at position

$$x = c.$$

Whenever the particle reaches the target, the trajectory terminates.

Tasks

1. Compute the distribution of first-passage times from lots of trajectories. Store each first-passage time in an array. Note that some trajectories won't reach the target – see section 3.5 in the lecture notes.
2. The single most likely value of a distribution is called its mode. From the diffusion law, it may seem reasonable that the mode is given by

$$t_{\text{mode}} = \alpha \frac{L^2}{D}$$

where L is the initial distance between walker and target, and α is a constant. (Note that the mean time here is infinite, which is why we focus on the mode time).

- Find α via simulations.
- Validate the above by calculating α analytically. That is, calculate the maximum of the first-passage time density in section 3.5 of the lecture notes.

Tip: consider the logarithm of the density for a cleaner calculation.

3. * Now consider diffusion inside a finite interval $B \leq x \leq c$. The boundary condition at B is reflecting.
4. * The target may itself diffuse around, with diffusion constant D_{target} . Redo tasks 1-2 and check what happens.
5. *